

Lectures

Stat 154/254: Statistical Machine Learning

Lectures marked with ★ are still in progress.

Introduction

- [Class introduction](#)
- [List of key concepts](#)

Unit 1: Regression in a machine learning setting

- [Linear regression in matrix notation \(review\)](#)
- [Linear regression in an ML setting](#)
- [Basis expansions and splines for linear regression](#)
- [L1 and L2 penalization](#)

Unit 2: Classification

- [Empirical risk minimization for classification](#)
- [Trading off false positives and negatives](#)
- [Perceptron algorithm \(bonus content\)](#)

Unit 3: Risk and complexity

- [Asymptotics and bias–variance tradeoff in linear regression](#)
- [Regret decomposition](#)
- [Cross validation](#)

- Held-out data and cross-validation
 - Simulated bias–variance tradeoff example
 - Cross-validation cautionary note
- Generalization bounds
 - Generalization error and uniform laws
 - Uniform laws for smooth function classes
 - VC dimension and zero–one loss

Unit 4: Trees and weak learners

- Classification and regression trees
- Bagging weak learners
- Boosting weak learners

Unit 5: Neural networks and optimization

- Optimization notes

Prerequisites

Mathematical statistics, probability, linear algebra, multivariate calculus, and Python are all prerequisites.

For assistance with mathematics prerequisites, the [Student Learning Center](#) provides additional assistance.

A summary of the required linear algebra skills can be found [here](#).